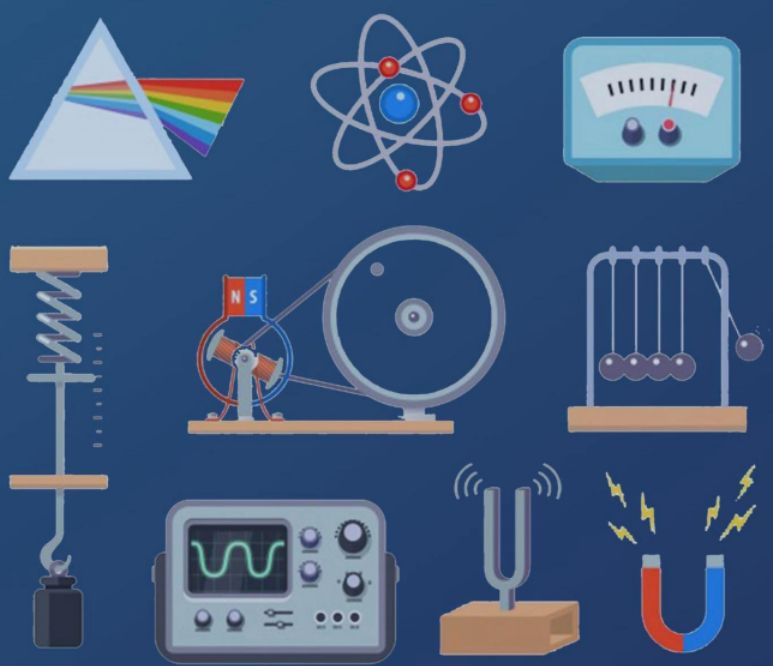
PHYSICS

Kinetic Theory of Gases (KTG)



SHEET

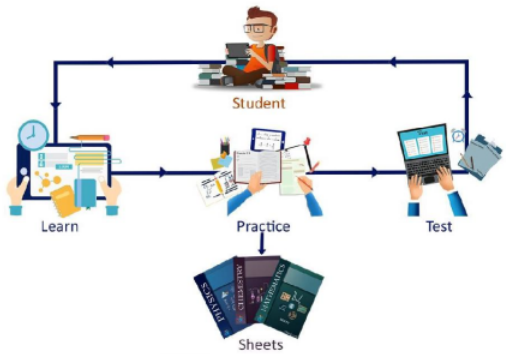
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Ideal Gas Equation - Part 1	1-5	1,2	1-5	1-3,6
2	Ideal Gas Equation - Part 2	6,7	3		4,7
3	Degree of Freedom	8,9	5,6	7, 9	5,8
4	Mean Free Path	10			
5	After Full Chapter	11	4	6, 8	

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Ideal Gas Equation - Part 1

Q1. Three containers of the same volume contain three different gases at same temp. The masses of the molecules are m_1, m_2 and m_3 and the number of molecules in their respective containers are N_1, N_2 and N_3. The gas pressure in the containers are P_1, P_2 and P_3 respectively. All the gases are now mixed and put in one of the containers. The pressure P of mixture will be :–

(A) $P < (P_1 + P_2 + P_3)$

(B) $P = \frac{P_1 + P_2 + P_3}{3}$

(C) $P = P_1 + P_2 + P_3$

(D) $P > (P_1 + P_2 + P_3)$

Q2. If rms velocity of a gas is V_{rms} = 1840 m/s and its density $\rho = 8.99 \times 10^{-2}$ kg/m^3, the pressure of the gas will be -

(A) 1.01 N/m^2

(B) 1.01×10^3 N/m^2

(C) 1.01×10^5 N/m^2

(D) 1.01×10^7 N/m^2

Q3. The volume of an ideal gas is V at pressure P and temperature T . The mass of each molecule of the gas is m. The density of gas will be :- (k is Boltzmann's constant)

(A) mkT (B) $\frac{Pm}{kT}$

(C) $\frac{P}{kTV}$ (D) $\frac{P}{kT}$

Q4. The number of oxygen molecules in a cylinder of volume 1 m^3 at a temperature of 27°C and pressure 13.8 Pa is (Boltzmaan's constant $k=1.38 \times 10^{-23}$ JK^{-1}) is :-

(A) 6.23×10^{26} (B) 0.33×10^{28}

(C) 3.3×10^{21} (D) none of these

Q5. The rms speed of an ideal diatomic gas at temperature T is v. When gas dissociates into atoms then its new rms speed becomes double. The temperature at which the gas dissociated into atoms was :-

(A) T (B) $\sqrt{2}$ T

(C) $\frac{T}{2}$ (D) 2T

Ideal Gas Equation - Part 2

Q6. **Statement–1 :** A real gas behaves as an ideal gas at high temperature and low pressure.

Statement–2 : Liquid state of an ideal gas is impossible.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Q7. The thermodynamic co-ordinates of a jar filled with gas A are P, V and T and another jar B filled with another gas are 2P, $\frac{V}{4}$ and 2T, where the symbols have their usual meaning. The ratio of the number of molecules of jar A to those of jar B is :–

(A) 4 : 1 (B) 2 : 1

(C) 1 : 2 (D) 1 : 1

Degree of Freedom

Q8. Mean kinetic energy (or average energy) per gram molecule of a monoatomic gas is given by :–

(A) $\frac{3}{2}RT$ (B) $\frac{1}{2}kT$

(C) $\frac{1}{2}RT$ (D) $\frac{3}{2}kT$

Q9. Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. The temperature of the mixture will be :

(A) $\frac{T_1+T_2}{n_1+n_2}$ (B) $\frac{T_1}{n_1}+\frac{T_2}{n_2}$

(C) $\frac{n_2T_1+n_1T_2}{n_1+n_2}$ (D) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$

Mean Free Path

Q10. If the pressure in a closed vessel is reduced by drawing out some gas, the mean free path of the molecules :-

(A) is decreased

(B) is increased

(C) remains unchanged

(D) increases or decreases according to the nature of the gas

After Full Chapter

Q11. Assume that the temperature remains essentially constant in the upper parts of the atmosphere. The atmospheric pressure varies with height as : (the mean molecular weight of air is M and P_0 = atomspheric pressure at ground reference)

(A) $P_0 e^{\frac{-3Mgh}{2RT}}$ (B) $P_0 e^{\frac{-Mgh}{2RT}}$

(C) $P_0 e^{\frac{-3Mgh}{RT}}$ (D) $P_0 e^{\frac{-Mgh}{RT}}$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. 28 gm of N_2 gas is contained in a flask at a pressure of 10 atm and at a temperature of 57°C. It is found that due to leakage in the flask, the pressure is reduced to half and the temperature reduced to 27°C. The quantity of N_2 gas that leaked out is :-

(A) 11/20 gm (B) 20/11 gm

(C) 5/63 gm (D) 63/5 gm

Q2. Relation PV = RT is given for following condition for real gas :–

(A) high temperature and high density

(B) low temperature and low density

(C) high temperature and low density

(D) low temperature and high density

Q3. A gas is enclosed in a vessel at a constant temperature at a pressure of 5 atmosphere and volume 4 litre. Due to a leakage in the vessel, after some time, the pressure is reduced to 4 atmosphere. As a result, the

(A) volume of the gas decreased by 20%

(B) average K.E. of gas molecule decreases by 20%

(C) 20% of the gas escaped due to the leakage

(D) 25% of the gas escaped due to the leakage

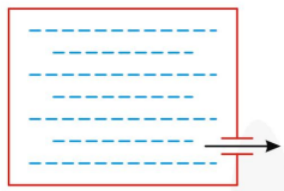
Q4. A vessel of volume V_0 contains an ideal gas at a pressure P_0. Gas is continuously pumped out of this vessel at a constant rate $\frac{dV}{dt} = r$, keeping the temperature constant. The pressure of the gas taken out equals the pressure inside the vessel. Find the pressure of the gas inside the vessel as a function of time.

(A) $2P_0 e^{-rt/V_0}$

(B) $3P_0 e^{-rt/V_0}$

(C) $-P_0 e^{-rt/V_0}$

(D) $P_0 e^{-rt/V_0}$



SUBJECTIVE TYPE

Q5. A cylinder of 200 litre capacity is containing H_2. The total average translational kinetic energy of molecules is 1.52×10^5 J.

The pressure of H_2 in the cylinder will be in N m^{-2} :-

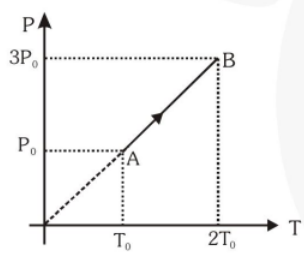
Q6. The ratio of average translational kinetic energy to rotational kinetic energy of a diatomic molecule at temperature T is :

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Pressure versus temperature graph of an ideal gas is shown in figure.Density of the gas at point A is ρ_0 . Density at B will be :-



(A) $\frac{3}{4}\rho_0$ (B) $\frac{3}{2}\rho_0$

(C) $\frac{4}{3}\rho_0$ (D) $2\rho_0$

Q2. At what temperature the V_{rms} of oxygen molecules will be equal to the escape velocity from the earths surface.

Given R = 8.3 J/K-mol &

escape speed = 11.2 km/s

(A) 1.61×10^5 K (B) 1.61×10^2 K

(C) 1.61×10^4 K (D) None of these

Q3. At temperature T, N molecules of gas A each having mass m and at the same temperature 2N molecules of gas B each having mass 2m are filled in a container. The mean square velocity of molecules of gas B is v^2 and mean square of x component of velocity of molecules of gas A is ω^2. The ratio of ω^2/v^2 is :

(A) 1 (B) 2

(C) 1/3 (D) 2/3

Q4. Three particles have speeds of 2u, 10u and 11u. Which of the following statements is correct ?

(A) The r.m.s. speed exceeds the mean speed by about u.

(B) The mean speed exceeds the r.m.s. speed by about u.

(C) The r.m.s. speed equals the mean speed.

(D) The r.m.s. speed exceeds the mean speed by more than 2u.

Q5. A container X has volume double that of container Y and both are connected by a thin tube. Both contains same ideal gas. The temperature of X is 200K and that of Y is 400K. If mass of gas in X is m then in Y it will be :

(A) m/8 (B) m/6

(C) m/4 (D) m/2

Q6. Two closed vessels of equal volume contain air at P_0 pressure, T_0 temperature and are connected through a narrow tube. If one of the vessel is now maintained at T_0 and other at T, what will be the pressure in the vessels.

(A) $\frac{2P_0}{T_0\left(\frac{1}{T}+\frac{1}{T_0}\right)}$ (B) $\frac{P_0}{T_0\left(\frac{1}{T}+\frac{1}{T_0}\right)}$

(C) $\frac{3P_0}{T_0\left(\frac{1}{T}+\frac{1}{T_0}\right)}$ (D) None of these

Q7. A vessel contains a mixture of one mole of oxygen and two moles of nitrogen at 300 K. The ratio of the average rotational kinetic energy per O_2 molecule to per N_2 molecule is :–

(A) 1 : 1 (B) 1 : 2

(C) 2 : 1

(D) depends on the moment of inertia of the two molecules

Q8. Air is filled at 60°C in a vessel of open mouth. The vessel is heated to a temperature T so that $\frac{1}{4}$th part of air escapes. The value of T is :–

(A) 171°C (B) 161°C

(C) 151°C (D) 181°C

Q9. Calculate the number of degrees of freedom of molecules of hydrogen in 1 cc of hydrogen gas at NTP.

(A) 1.344×10^{20}

(B) 1.344×10^{19}

(C) 1.344×10^{18}

(D) 1.344×10^{17}

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. During experiment, an ideal gas is found to obey a condition P^2/ρ= constant [ρ = density of the gas]. The gas is initially at temperature T, pressure P and density ρ. The gas expands such that density changes to $\rho/2$.

(A) The pressure of the gas changes to $\sqrt{2}$ P

(B) The temperature of the gas changes to $\sqrt{2}$ T

(C) The graph of the above process on the P–T diagram is parabola

(D) The graph of the above process on the P–T diagram is hyperbola

Q2. If the pressure of a gas is doubled at constant temperature, then the mean square velocity will become :-

(A) no change (B) double

(C) four times (D) none of the above

Q3. A vessel is partitioned in two equal halves by a fixed diathermic separator. Two different ideal gases are filled in left (L) and right (R) halves. The rms speed of the molecules in L part is equal to the mean speed of molecules in the R part. Then the ratio of the mass of a molecule in L part to that of a molecule in R part is :-



(A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\pi/4}$

(C) $\sqrt{2/3}$ (D) $3\pi/8$

Q4. A thin tube of uniform cross-section is sealed at both ends. It lies horizontally, the middle 5 cm containing mercury and the parts on its two sides containing air at the same pressure p. When the tube is held at an angle of 60° with the vertical, the length of the air column above and below the mercury pallet are 46 cm and 44.5 cm respectively. Calculate the pressure p in centimetres of mercury. The temperature of the system is kept at 30°C.

(A) 75.39 cm

(B) 54.39 cm

(C) 64.39 cm

(D) 85.39 cm

Q5. The temperature of an ideal gas consisting of rigid diatomic molecules is T = 300K. Calculate the angular root mean square velocity of a rotating molecule if its moment of inertia is equal to $I = 2.1\times10^{-39}\ g\,cm^2$.

(A) 6.3×10^{12} rad / sec

(B) 6.8×10^{12} rad / sec

(C) 3.6×10^{12} rad / sec

(D) 3.2×10^{12} rad / sec

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q6. At a temperature T K, the pressure of 4.0g argon in a bulb is p. The bulb is put in a bath having temperature higher by 50K than the first one. 0.8g of argon gas had to be removed to maintained original pressure. The temperature T is equal to

SUBJECTIVE TYPE

Q7. A vessel of volume $8.3 \times 10^{-3}\ m^3$ contains an ideal gas at temperature 27°C and pressure 200 kPa. The gas is allowed to leak till the pressure falls to 100 kPa and temperature increases to 327°C. What is the amount of gas in moles will be leaked out?

Q8. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T. Neglecting all vibrational modes, the total internal energy of the system is :-

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	B	C	D	B	A	A	D	B
11									
D									

Ex-1A

1. D **2.** C **3.** C **4.** D **5.** 5×10^5 N/m^2

6. 3/2

Ex-2								
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
B	A	D	A	C	A	A	A	A

Ex-2A

1. B,D **2.** A **3.** D **4.** A **5.** A

6. 200 K **7.** $\frac{1}{2}$ mole **8.** 11RT